Module 4: Lebesgue Measure

Measure axioms, continuity, outer measure, Carathéodory's criterion, and Lebesgue measurability

1. Measures

1.1 Definition and classical examples

On a measurable space $(\Omega, \mathcal{F})$, a *measure* is a function $\mu : \mathcal{F} \to [0, \infty]$ satisfying

$$\boxed{\mu(\varnothing) = 0, \qquad \mu\left(\bigsqcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} \mu(A_n) \quad (A_n \in \mathcal{F}).}$$

Here $\bigsqcup$ denotes disjoint union. The triple $(\Omega, \mathcal{F}, \mu)$ is a *measure space*; $+\infty$ is allowed.

- **Counting measure:** on $(\Omega, \mathcal{P}(\Omega))$, $\mu(A) = |A|$ if A is finite, and $+\infty$ otherwise.
- **Dirac measure:** for $x_0 \in \Omega$, $\delta_{x_0}(A) = \mathbf{1}_A(x_0)$ on any $\mathcal{F}$.
- **Probability measure:** a measure $\mathbb{P}$ with $\mathbb{P}(\Omega) = 1$.

1.2 Monotonicity and countable subadditivity

Monotonicity. If $A \subseteq B$ are measurable, then

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A).$$

Countable subadditivity. For arbitrary measurable A_n, disjointness is not needed for

$$\boxed{\mu\left(\bigcup_{n=1}^{\infty} A_n\right) \leq \sum_{n=1}^{\infty} \mu(A_n).}$$

Proof idea. Disjointify via $B_1 = A_1$ and $B_n = A_n \setminus \bigcup_{k<n} A_k$: then $\bigcup_n B_n = \bigcup_n A_n$ and $B_n \subseteq A_n$.

1.3 Finite and sigma-finite measures

A measure is *finite* if $\mu(\Omega) < \infty$. It is *sigma-finite* if $\Omega = \bigcup_{n\geq 1} E_n$ for measurable E_n with $\mu(E_n) < \infty$. Every finite measure is sigma-finite, but the converse fails. Lebesgue length on $\mathbb{R}$ is sigma-finite, since

$$\mathbb{R} = \bigcup_{n\geq 1} [-n, n], \qquad \lambda([-n, n]) = 2n < \infty,$$

although $\lambda(\mathbb{R}) = \infty$. Counting measure on an uncountable set is not sigma-finite.

Key point. A measure is countably *additive* on disjoint sets and countably *subadditive* on arbitrary measurable sets. These are distinct assertions.

2. Continuity of Measures

2.1 Continuity from below and from above

For increasing measurable sets $A_n \uparrow A$, meaning $A_n \subseteq A_{n+1}$ and $A = \bigcup_n A_n$, one has

$$\boxed{\mu(A_n) \uparrow \mu(A), \qquad \mu\left(\bigcup_n A_n\right) = \lim_{n\to\infty} \mu(A_n).}$$

Proof. Put $B_1 = A_1$, $B_n = A_n \setminus A_{n-1}$ $(n \geq 2)$. These are disjoint and $A_m = \bigsqcup_{n\leq m} B_n$, $A = \bigsqcup_{n\geq 1} B_n$; apply countable additivity.

For decreasing sets $A_n \downarrow A$, meaning $A_n \supseteq A_{n+1}$ and $A = \bigcap_n A_n$, we have

$$\boxed{\mu(A_1) < \infty \quad \Longrightarrow \quad \mu(A_n) \downarrow \mu(A).}$$

Proof. Set $B_n = A_1 \setminus A_n \uparrow A_1 \setminus A$. Continuity from below applies to B_n. Since $\mu(A_1) < \infty$, subtraction in $\mu(A_n) = \mu(A_1) - \mu(B_n)$ is legitimate, and taking limits gives $\mu(A)$.

Why the finiteness condition matters. For Lebesgue measure, let $A_n = [n, \infty)$. Then $A_n \downarrow \varnothing$ but

$$\boxed{\mu(A_n) = \infty \text{ for every } n, \qquad \mu\left(\bigcap_n A_n\right) = 0.}$$

Every point eventually leaves these infinite-length tails; subtraction would require the undefined $\infty - \infty$.

2.2 Connection with set limsup and liminf

For arbitrary sets, recall the Module 1 definitions

$$\limsup_n A_n = \bigcap_N \bigcup_{n\geq N} A_n, \qquad \liminf_n A_n = \bigcup_N \bigcap_{n\geq N} A_n.$$

For monotone sequences, $\liminf A_n = \limsup A_n$ equals $\bigcup_n A_n$ (increasing) or $\bigcap_n A_n$ (decreasing). Measure continuity preserves these limits, with finiteness from above; it is an *order-sequential*, not unrestricted, continuity.

Counterexample without monotonicity. For $A_n = [n, n+1]$, $\liminf A_n = \limsup A_n = \varnothing$, yet $\lambda(A_n) = 1$ for all n.

Key point. Continuity means compatibility with limits. Countable additivity implies continuity from below; finite mass prevents a loss of mass "at infinity" when sets decrease.

3. Outer Measure

3.1 Constructing Lebesgue outer measure

How can interval length be extended to general sets? Cover the target set by countably many open intervals and minimize the total covering length. For every $A \subseteq \mathbb{R}$, define

$$\boxed{\lambda^*(A) = \inf\left\{\sum_{n=1}^{\infty} |I_n| : A \subseteq \bigcup_{n=1}^{\infty} I_n,\ I_n \text{ open intervals}\right\}.}$$

The infimum need not be attained; λ^* is defined on *all* subsets of $\mathbb{R}$. An *outer measure* $\mu^* : \mathcal{P}(\Omega) \to [0,\infty]$ satisfies

$$\boxed{\mu^*(\varnothing) = 0, \quad A \subseteq B \Rightarrow \mu^*(A) \leq \mu^*(B), \quad \mu^*\left(\bigcup_n A_n\right) \leq \sum_n \mu^*(A_n).}$$

For λ^*, monotonicity follows by reusing covers; subadditivity follows by combining near-optimal covers.

3.2 Basic examples

Intervals. For $a < b$, $\lambda^*([a,b]) = b - a$. An interval $(a-\varepsilon, b+\varepsilon)$ gives the upper bound $b - a + 2\varepsilon$. For the lower bound, compactness gives a finite subcover; a chain of overlapping intervals from a to b shows its lengths sum to at least $b - a$. The original covering sum is no smaller.

Countable sets. If $A = \{x_1, x_2, \ldots\}$, cover x_n by an open interval of length $\varepsilon/2^n$. The total length is ε, so $\lambda^*(A) = 0$. In particular,

$$\boxed{\lambda^*(\mathbb{Q} \cap [0,1]) = 0, \qquad \overline{\mathbb{Q} \cap [0,1]} = [0,1].}$$

Thus topological density does not imply positive measure.

3.3 Why outer measure is not yet a measure

Outer measure guarantees only *subadditivity*; even for disjoint A, B it need not satisfy $\lambda^*(A \cup B) = \lambda^*(A) + \lambda^*(B)$. If it were a countably additive measure on all $\mathcal{P}(\mathbb{R})$, its translation invariance and correct interval lengths would contradict the Vitali-set argument from Module 3. The remedy is to *restrict the domain* to sets on which splitting is additive.

Key point. The covering construction assigns an outer size to every set, but the full power set is too large a domain for a translation-invariant, countably additive extension of interval length.

4. Carathéodory's Criterion

4.1 Measurability by additive splitting

Let μ^* be an outer measure on Ω. A set $A \subseteq \Omega$ is *Carathéodory measurable* if for *every* $E \subseteq \Omega$,

$$\boxed{\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \setminus A).}$$

The inequality "$\leq$" follows automatically from subadditivity; the criterion demands the reverse inequality as well. Thus a measurable A splits any test set without increasing outer size; testing only $E = A$ would be vacuous.

Null sets are measurable. If $\mu^*(A) = 0$, then $\mu^*(E \cap A) = 0$ and $\mu^*(E \setminus A) \leq \mu^*(E)$. Together with subadditivity this proves the criterion. Every subset of an outer-null set is likewise measurable.

4.2 Carathéodory's theorem

Theorem. Let μ^* be any outer measure. Then

$$\boxed{\begin{array}{l}\mathcal{M} = \{A \subseteq \Omega : A \text{ satisfies the splitting criterion}\} \\ \quad \text{is a sigma-algebra, and} \qquad \mu = \mu^*|_{\mathcal{M}} \text{ is a measure.}\end{array}}$$

Proof sketch. The empty set is measurable, and complementation just exchanges the two pieces of a split. Repeated splitting establishes finite unions. For pairwise disjoint measurable A_n, repeated application of the criterion, followed by countable subadditivity, shows that $A = \bigcup_n A_n$ is measurable and

$$\mu^*(A) = \sum_n \mu^*(A_n).$$

Disjointification extends this to arbitrary countable unions.

Key point. Carathéodory's test converts a universally defined, merely subadditive outer measure into a genuinely countably additive measure on a sigma-algebra.

5. Lebesgue Measure

5.1 Definition and main properties

Apply Carathéodory's theorem to Lebesgue outer measure λ^*:

$$\boxed{\begin{array}{c}\mathcal{L} = \{A \subseteq \mathbb{R} : A \text{ is Carathéodory measurable for } \lambda^*\}, \\ \lambda = \lambda^*|_{\mathcal{L}}, \qquad (\mathbb{R}, \mathcal{L}, \lambda).\end{array}}$$

This is *Lebesgue measure*, with the following properties:

- **Length:** $\lambda([a,b]) = b - a$; hence $\lambda(\mathbb{R}) = \infty$.
- **Translation invariance:** $A \in \mathcal{L} \Rightarrow A + t \in \mathcal{L}$ and $\lambda(A + t) = \lambda(A)$.
- **Countable additivity:** $\lambda(\bigsqcup_n A_n) = \sum_n \lambda(A_n)$.
- **Completeness:** every subset of a Lebesgue null set is Lebesgue measurable and null.

Measure-theoretic completeness is *not* Cauchy completeness from metric-space theory.

5.2 Borel sets and Lebesgue measure

Every open subset of $\mathbb{R}$ is Lebesgue measurable (proved by splitting interval covers, first for intervals and then for open sets). Since $\mathcal{L}$ is a sigma-algebra containing the usual open sets,

$$\boxed{\mathcal{B}(\mathbb{R}) = \sigma(\mathcal{T}_{\text{usual}}) \subseteq \mathcal{L}.}$$

The Borel sigma-algebra is generated by topology; Lebesgue measurability is determined by additive splitting of outer measure.

5.3 Borel versus Lebesgue measurability

Both inclusions are strict:

$$\boxed{\mathcal{B}(\mathbb{R}) \subsetneq \mathcal{L} \subsetneq \mathcal{P}(\mathbb{R}).}$$

Cantor set (middle-thirds construction). Start with $C_0 = [0,1]$; at each step remove the open middle third of every remaining interval:

$$C_1 = [0, \tfrac{1}{3}] \cup [\tfrac{2}{3}, 1], \qquad C = \bigcap_{n=0}^{\infty} C_n.$$

Each C_n is a union of 2^n closed intervals of length 3^{-n}. Thus C is closed (hence compact and Borel), and

$$\boxed{0 \le \lambda(C) \le \lambda(C_n) = (\tfrac{2}{3})^n \to 0, \qquad \lambda(C) = 0.}$$

Yet C is *uncountable*: its elements are exactly the numbers with ternary expansions using only 0 and 2, so $|C| = 2^{\aleph_0}$. This is an uncountable compact set of measure zero.

Why the inclusions are strict. Every $S \subseteq C$ is Lebesgue measurable by completeness. But $|\mathcal{P}(C)| = 2^{2^{\aleph_0}} > 2^{\aleph_0} = |\mathcal{B}(\mathbb{R})|$, so some such S cannot be Borel. A Vitali set is not Lebesgue measurable, proving the second strict inclusion.

Key point. The construction is *interval length → outer measure → Carathéodory measurability → Lebesgue measure*. The restriction of the domain restores countable additivity.